VocalTuneBtw manual

Real-time pitch correction for monophonic vocals, built natively for Linux.

Version 2.0.5. To install it, see the README.

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Getting started

1. Put VocalTuneBtw on a track with a **single voice**. It's made for monophonic vocals, not choirs or doubled parts.
2. Set **Key** and **Scale** to your song's.
3. Open the settings behind the gear icon and choose the **Voice Range** that fits the singer.
4. Click **Natural** for transparent correction or **Hard** for the robotic snap.



The controls

Main view

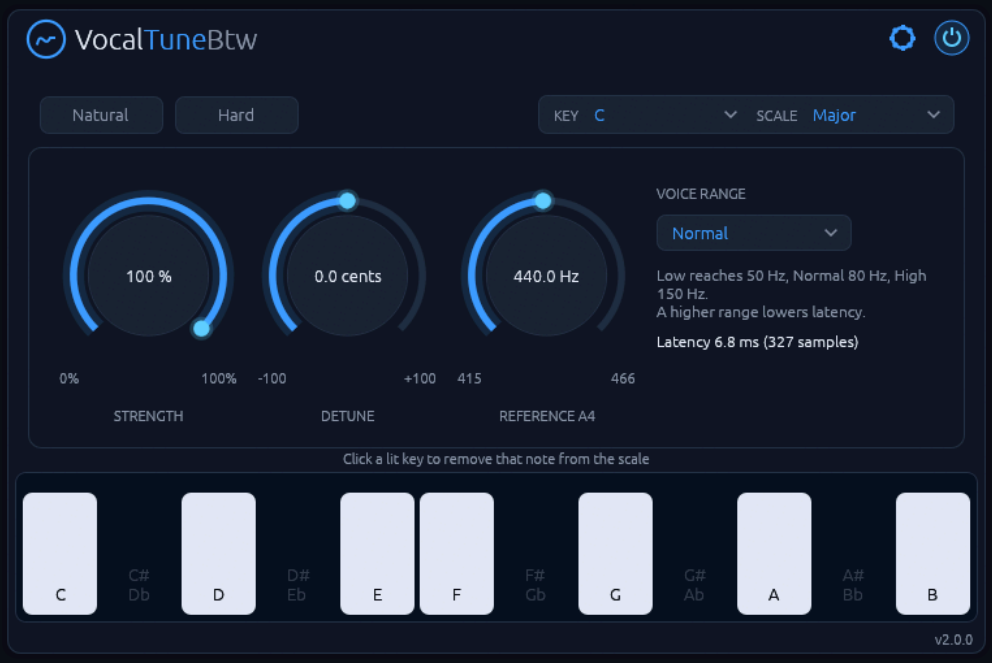
Control	Range	What it does
Key / Scale	12 roots × 14 scales	The notes the voice is pulled to.
Keyboard	per note	One octave. Lit keys are in the scale; the note being corrected is blue. Click a key in the scale to switch it off (dark) or on again.

Control	Range	What it does
Retune Speed	1 to 200 ms	How fast the voice moves to the target. Low values give the hard, stepped effect.
Humanize	Off to 100%	Deviations smaller than this are left uncorrected (0 to 50 cents).
Natural / Hard	presets	Set Retune Speed and Humanize in one click. The active preset lights up.
Power	On / Off	Bypass with a 10 ms crossfade. The switch is blue while the effect runs; while bypassed it turns grey and the tuning meter dims.

Settings

Behind the gear icon:

Control	Range	What it does
Strength	0 to 100%	How much of the pitch error is corrected.
Detune	±100 cents	Shifts every target note by a fixed amount.
Reference A4	415 to 466 Hz	Concert pitch the scale is tuned to.
Voice Range	Low / Normal / High	The singer's range. Sets both the detection range and the latency.



All controls are host parameters and can be automated. Clicking the logo opens the plugin's page.

How to use it

Natural or Hard

- **Natural** sets Retune Speed to 50 ms with light Humanize: the voice is guided to the note and keeps its vibrato and slides.
- **Hard** sets Retune Speed to 1 ms with no Humanize: the voice jumps between notes, which is the stepped, robotic effect.

Between these two, Retune Speed is the main control. Start from a preset and move it by ear.

Humanize

Humanize leaves alone anything already close to the target, so held notes keep their natural movement. Raise it if long notes sound too flat and still. Lower it if notes stay audibly off.

Choosing the notes

The scale decides which notes the voice may land on. Choose **Chromatic** to correct to the nearest semitone without choosing a key. **Custom** and the keyboard let you take notes away: click a lit key to remove that note, and the voice snaps to the neighbouring notes instead. Click it again to bring it back. This helps when a singer keeps landing on a note you don't want.

The scales are Chromatic, Major, Natural, Harmonic and Melodic Minor, Dorian, Phrygian, Lydian, Mixolydian, Locrian, Major and Minor Pentatonic, Blues and Custom.

Strength, Detune and Reference

Strength below 100% corrects only part of the error, for a subtle touch. **Detune** shifts every target by a fixed amount, up to ±100 cents. **Reference A4** tunes the whole scale, for material that isn't at 440 Hz.

Voice Range and latency

The pitch shifter needs about half a cycle of the lowest note it has to handle, so a higher Voice Range means lower latency. Pick the **highest range that still covers the singer's lowest note**. Values at 48 kHz:

Voice Range	Lowest note	Latency
Low	50 Hz	10.8 ms
Normal	80 Hz	6.8 ms
High	150 Hz	3.8 ms

The current value is shown on the settings page, and the plugin reports it to the host for automatic delay compensation.

Good to know

- **The start of a note:** detection needs some audio before it recognises a new note, so the start of a note is corrected roughly 25 to 30 ms late. This doesn't delay the audio itself.
- **Stereo:** the channels are shifted together, so the image stays intact.
- **Audio:** mono or stereo, at whatever sample rate the host uses.
- **Hosts:** tested in Bitwig Studio. Other CLAP or VST3 hosts should work but haven't been tried.
- **Display:** X11, or XWayland on a Wayland desktop, with a Vulkan driver.

Troubleshooting

- **The plugin doesn't show up:** check that the bundle is in `~/.clap`, `~/.vst3`, `/usr/lib/clap` or `/usr/lib/vst3`, then rescan plugins in your DAW.

- **The editor window stays black or doesn't open:** check that a Vulkan driver is installed. The `vulkaninfo` tool from `vulkan-tools` should list your GPU. Bitwig Studio writes plugin errors to `~/.BitwigStudio/log/engine.log`.